Trends in Biochemical Sciences

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Spotlight

All Ribosomes Are Created Equal. Really?

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Ribosomes are generally thought of as molecular machines with a constitutive rather than regulatory role during protein synthesis. A study by Slavov *et al.* [1] now shows that ribosomes of distinct composition and functionality exist within eukaryotic cells, giving credence to the concept of 'specialized' ribosomes.

The 60 years since the first description of the ribosome [2] have seen enormous progress in our understanding of how these intricate molecular machines carry out protein synthesis [3]. Protein synthesis, or mRNA translation, and its key aspects are conserved throughout evolution. Accordingly, the core features of the ribosome have been preserved, including most ribosomal RNA (rRNA) and protein (RP) components, as well as the catalytic mechanism of peptide bond formation that harks back to a primordial RNA world. A major feat at the turn of the century was to obtain high-resolution structures of the bacterial ribosome by X-ray crystallography. This has since been followed by multiple ribosome structures of prokaryotic and eukaryotic origin, representing different sub-steps of the translation cycle [3].

The universal requirement of protein synthesis for life, and perhaps also the need to work with homogenous particles in structural biology, have led to an expectation that ribosomes of a given species will be of a fixed composition and operate in a default fashion to translate the genetic code in mRNA. Furthermore, the realization that gene expression is commonly regulated at the post-transcriptional level, including the control of translation, put much of the spotlight on *cis*-acting elements within the mRNA template. These elements interact with *trans*-acting RNA-binding proteins, microRNAs and eukaryotic initiation factors to affect ribosome recruitment in a highly combinatorial manner [4,5]. Nonetheless, there have also been clues for an active role of the ribosome in shaping a specific spatio-temporal outcome of protein synthesis, as heralded in [6] and reviewed in [7,8].

The typical eukaryotic ribosome consists of four rRNAs and 80 RPs. Some of this inventory was added at the transition from prokaryotes to eukaryotes; for example, there are marked increases in the rRNA 'expansion segments' and several eukaryote-specific RPs. Additionally, there is considerable diversity in the expression of RP paralogs among eukaryotes. Divergent RP mutant phenotypes in model species, but also the highly selective pathologies seen in 'ribosomopathies' (genetic diseases involving ribosomal defects [9]), were among the key observations that led to the proposal of 'specialized' ribosomes. Within a given organism, ribosomes of distinct composition might be seen during development and between differentiated tissues, in response to extrinsic cues, or among spatially or functionally distinct intracellular pools (Figure 1A). Cells might use several means to 'build' ribosomes with special properties (Figure 1B). For instance, ribosomes can associate with potentially sub-stoichiometric auxiliary factors to provide an interface with other cellular processes. In addition, several RPs are known to carry post-translational modifications (PTMs); for example, phosphorylation of the RP S6 has long been under investigation for a regulatory function. Similarly, exquisitely precise patterns of certain rRNA nucleoside modifications have been recorded, as has the incorporation of variant rRNA sequences. The 'swapping-in' of RP paralogs, or a selective loss of canonical RPs, are further options reviewed in [7,8].

A key challenge is to obtain direct experimental proof for ribosomes of distinct composition from an *in vivo* context and furthermore to link such heterogeneity to separate (molecular) functions. Slavov *et al.* [1] tackled the issue using mass spectrometry to analyse ribosomes purified from two divergent eukaryotes: budding yeast (*Saccharomyces cerevisiae*) and mouse embryonic stem cells (ESCs). In addition to subjecting cells to different growth conditions they further fractionated cell extracts by sucrose density gradient ultracentrifugation. This technique separates translation complexes by the number of ribosomes lined up on the same mRNA molecule. Their key finding was that the stoichiometry of core RPs present in ribosomes depended on the type of translation complex. Some RPs were enriched in monosomes (ribosomes sedimenting as a single entity); others were enriched in polysomes (multiple ribosomes jointly translating an mRNA molecule) (Figure 1C).

Biased RP-stoichiometry patterns were apparent in both yeast and ESCs, and there was conservation with regard to the RPs affected. Interestingly, levels of RPs buried within the ribosome structure tended to be invariant, whereas those on the ribosome surface changed the most. The patterns could not be explained by the RP incorporation order during ribosome biogenesis, the presence of partially 'built' ribosomes, or a simple RP paralog swap. Effects of RP PTMs were also ruled out, as was selective ribosomal subunit loss during centrifugation. Since monosomes and lower-order polysomes shared similar composition they also were not simply distinguishing between inactive and active ribosomes. Furthermore, both in yeast and in ESCs, they could draw on cellular fitness data when individual RPs were deleted to examine functional differences between preferentially polysomal and monosomal RPs. In both cases, mutant cell fitness was inversely proportional to the degree of RP enrichment with polysomes. In yeast, they further

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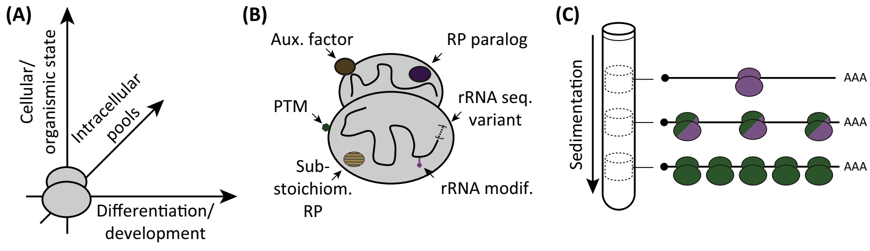

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Figure 1. The Concept of 'Specialized' Ribosomes. (A) Organisms might produce ribosomes of distinct composition and specialization as a function of development or tissue differentiation, in response to changes in cellular or organismic state, or to serve distinct functions within separate intracellular pools. (B) Cellular toolkit to create specialized ribosomes. Abbreviations: RP, ribosomal protein; PTM, post-translational modification. (C) Principle of sucrose density gradient ultracentrifugation and visualization of main finding by Slavov *et al.* [1]. Ribosomes are separated based on the number of mRNAs they jointly translate. Ribosomes with different degree of mRNA association have distinct RP stoichiometry (indicated by change in color).

demonstrated that polysome-enrichment of RPs correlated with the degree of RP gene induction when cell growth is stimulated.

The study by Slavov *et al.* [1] has thus made substantial progress in characterising 'specialized' ribosomes. Important questions remain unanswered or have been newly raised by this work. For instance, how is selective incorporation of RPs achieved? How many distinct types of ribosomes exist within a cell or organism? How do specialized ribosomes carry out their selective functions? Are they adapted for the translation of different pools of mRNAs and what are the mRNA features that cooperate in this matchmaking between ribosomes and mRNA? Recent findings that translation of mRNAs encoding the homeobox developmental regulators is dependent on ribosomes containing the RP L38 are offering initial clues in this respect [7]. More broadly, what can we learn from this about human diseases such as the ribosomopathies and cancer? Slavov *et al.* [1] noted that monosome-enriched RPs are frequently mutated in cancer, whereas RPs whose overexpression promotes cancer are enriched in polysomes. Can specialized ribosomes thus promote 'oncogenic' protein synthesis and cell transformation?

We can expect that active research in this field will provide us with further thought provoking results in the coming years.

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